

# Isobutane (iso-C<sub>4</sub>H<sub>10</sub>)

*A flammable, colorless liquefied gas.*

## GRADES & SPECIFICATIONS

| Grade                | % Purity | H <sub>2</sub> O | O <sub>2</sub> | Other HC | Sulfur | N <sub>2</sub> |
|----------------------|----------|------------------|----------------|----------|--------|----------------|
| Research             | 99.99%   | ≤2               | ≤5             | ≤90      | ≤1     | ≤20            |
| Instrument           | 99.5%    | ≤3               | ≤100           | ≤4,500   | ≤1     | ≤400           |
| Chemically Pure (CP) | 99.0%    | —                | —              | —        | —      | —              |

Concentrations given are ppm by volume unless otherwise specified.  
Individual or Batch Certificate of Analysis available upon request.

## ORDERING INFORMATION

| Product              | Cyl. Size | Contents (lbs) | Valve (CGA) | Product Number | Pressure @70°F (psig) | Delivery Range (psig) |
|----------------------|-----------|----------------|-------------|----------------|-----------------------|-----------------------|
| Research             | 350       | 111            | 510         | BI-RES-350     | 31                    | 0–25                  |
|                      | LP5       | 22             | 510         | BI-RES-LP5     | 31                    |                       |
|                      | LP1       | 5              | 510         | BI-RES-LP1     | 31                    |                       |
| Instrument           | 1/2 TON   | 466            | 510         | BI-INST-12TON  | 31                    | Single-stage          |
|                      | 350       | 111            | 510         | BI-INST-350    | 31                    |                       |
|                      | LP5       | 22             | 510         | BI-INST-LP5    | 31                    |                       |
| Chemically Pure (CP) | 1/2 TON   | 466            | 510         | BI-CP-12TON    | 31                    | Single-stage          |
|                      | 350       | 111            | 510         | BI-CP-350      | 31                    |                       |
|                      | LP5       | 22             | 510         | BI-CP-LP5      | 31                    |                       |

## TECHNICAL DATA & SHIPPING INFORMATION

|                     |                             |                     |                     |
|---------------------|-----------------------------|---------------------|---------------------|
| Molecular Weight    | 58.1222 g/mol               | U.S. DOT Name       | Isobutane           |
| Specific Volume     | 6.66 cf/lb @ 70°F & 1 ATM   | ID Number           | UN 1969             |
| Boiling Point       | –11.7 °C                    | DOT Hazard Class    | 2.1 — Flammable Gas |
| Flammability in Air | Flammable — 1.8–8.4% in Air | CAS Registry Number | 75-28-5             |

WSM offers a wide range of gas supply modes, including high-pressure cylinders, liquid cylinders (dewars), MicroBulk and bulk. Your WSM specialty gas representative can help you select the option best suited to your operation, including automatic changeover systems for uninterrupted supply.

**WSM Quality Policy.** The purpose of the WSM Quality System is to continually improve our manufacturing and related processes to provide our customers with the highest product purity, consistency, and service.